Cheap at every layer: hardware, connectivity, power, data, and coverage.

- **~€24 BOM** per node at volume
- **Install in 5 minutes:** stake, register, joins the mesh
- **Connectivity ~€1 per node** (only gateways use WiFi)
- **Self-powered:** solar + LiPo, no grid, no battery swaps
- **Tiny energy budget:** deep-sleep ~7 µA, ~60 mAh/year
- **Free weather data** (Open-Meteo / DWD)
- **A few hundred sensors cover all 126k trees**
- **Rewards are existing city capacity,** not cash

A multi-truck route planner, on a live per-tree model of the city.

- **Multi-truck route planning,** ranked by water stress
- **Risk triage:** young, high-stress trees first
- **Turn-by-turn** for citizens to the nearest tree
- **A live per-tree, per-street model** underneath
- **Proof an intervention worked:** before/after on real data
- **Evidence** for compliance and budget decisions
- **Closed loop:** plan, act, verify

The probe reads the top 5 cm. Physics models the rest of the tree, and most of the city.

- **The probe reads only the top ~5 cm** of soil
- **Roots drink 20–30 cm down**
- **Physics models unmeasured trees** from forecast + nearby sensors
- **A few sensors cover many trees**
- **Young trees: dense sensors** (high risk, citizen-verifiable)
- **Old trees: sparse** (deep roots, they survive)
- **Honest:** physics estimates, calibrates as sensors deploy